

SegNet4D: Efficient Instance-Aware 4D Semantic Segmentation for LiDAR Point Cloud

Neng Wang¹, Ruibin Guo¹, Chenghao Shi¹, Ziyue Wang, Hui Zhang, Huimin Lu², *Member, IEEE*,
Zhiqiang Zheng, and Xieyuanli Chen¹, *Member, IEEE*

Abstract—4D LiDAR semantic segmentation classifies the semantic category of each LiDAR point and detects whether it is dynamic, a critical ability for tasks like obstacle avoidance and autonomous navigation. Existing approaches often rely on computationally heavy 4D convolutions or recursive networks, which result in poor real-time performance. In this paper, we introduce SegNet4D, a novel real-time 4D semantic segmentation network, offering both efficiency and strong semantic understanding. SegNet4D addresses 4D segmentation as two tasks: single-scan semantic segmentation and moving object segmentation, each tackled by a separate network head. Both results are combined in a motion-semantic fusion module to achieve comprehensive 4D segmentation. Additionally, instance information is extracted from the current scan and exploited for instance-wise segmentation consistency. Extensive experiments on the SemanticKITTI and nuScenes datasets demonstrate that our method outperforms the state-of-the-art in both 4D semantic segmentation and moving object segmentation. Through detailed runtime analysis, our method shows greater efficiency, enabling real-time operation. Besides, its effectiveness and efficiency have also been validated on a real-world robotic platform. The implementation of our method has been released at <https://github.com/nubot-nudt/SegNet4D>

Note to Practitioners—This paper focuses on enhancing the efficiency of 4D semantic segmentation for applications in autonomous driving or navigation. Existing LiDAR-based 4D semantic segmentation methods fall short of real-time processing capabilities, thereby severely limiting their practicality for autonomous vehicles and robotic systems. To tackle these issues, we design a high-efficiency 4D semantic segmentation network that not only performs real-time operations on real-world robotic systems but also delivers superior performance, validating its practical utility. Future work can further leverage the instance information we introduced to improve the network's functionality by achieving panoptic segmentation or 4D panoptic segmentation. Furthermore, some studies may use the 4D semantic labels predicted by our approach to reinforce tasks associated with robotic autonomy.

Index Terms—4D semantic segmentation, deep learning, moving object segmentation, LiDAR point cloud.

I. INTRODUCTION

LIGHT Detection and Ranging (LiDAR) sensors, have become important for autonomous vehicles and robots to achieve autonomy due to their resilience to illumination changes, wide field of views, and accurate range measurements [1], [2]. As a result, LiDAR-based semantic perception has garnered extensive research interest in recent years. Semantic segmentation using 3D LiDAR data aims to assign a specific semantic class label to each point in the acquired point clouds, serving as a fundamental task in 3D perception and scene understanding. This technology significantly enhances key autonomous functions such as navigation [3], point cloud registration [4], [5], and simultaneous localization and mapping (SLAM) [6], [7].

In practical applications, however, single-scan 3D LiDAR semantic segmentation methods [1], [8], [9], [10] face inherent limitations due to their inability to capture motion dynamics, a crucial factor for effective obstacle avoidance and local path planning in autonomous systems. To address this, 4D semantic understanding is necessary. As shown in Fig. 1, it categorizes each LiDAR point while identifying its dynamic attributes [11], [12], [13]. This task typically relies on sequential LiDAR scans to capture motion information. In this regard, some methods directly stack historical LiDAR scans into a single point cloud and feed it into a single-scan-based semantic network for achieving 4D segmentation [1], [9], [14], which leads to suboptimal performance due to the lack of motion information association. Other approaches use 4D convolutional neural networks [15] or recursive networks [12], [16] to extract motion features from input sequential LiDAR scans. However, these approaches impose heavy computational burdens and make real-time operations challenging. Moreover, existing methods commonly treat 4D semantic segmentation in an end-to-end fashion, directly predicting semantic labels for all categories, including those moving and static classes. This may lead to limited performance in recognizing dynamic objects because moving points are often fewer than static background points in most datasets.

To address these challenges, building upon our previous work InsMOS [17], we propose an extended framework for 4D semantic segmentation. Our key idea is to tackle 4D semantic segmentation by dividing it into two subtasks: single-scan semantic segmentation (SSS) and moving object

Received 24 December 2024; revised 1 April 2025; accepted 2 May 2025. Date of publication 9 May 2025; date of current version 23 May 2025. This article was recommended for publication by Associate Editor Y. Sun and Editor J. Yi upon evaluation of the reviewers' comments. This work was supported in part by the National Science Foundation of China under Grant 62403478, Grant U22A2059, and Grant 62203460; in part by the Young Elite Scientists Sponsorship Program by China Association for Science and Technology (CAST) under Grant 2023QNRC001; and in part by the Major Project of Natural Science Foundation of Hunan Province under Grant 2021JC0004. (*Corresponding author: Xieyuanli Chen.*)

The authors are with the College of Intelligence Science and Technology and the National Key Laboratory of Equipment State Sensing and Smart Support, National University of Defense Technology, Changsha 410073, China (e-mail: xieyuanli.chen@nudt.edu.cn).

Digital Object Identifier 10.1109/TASE.2025.3568001

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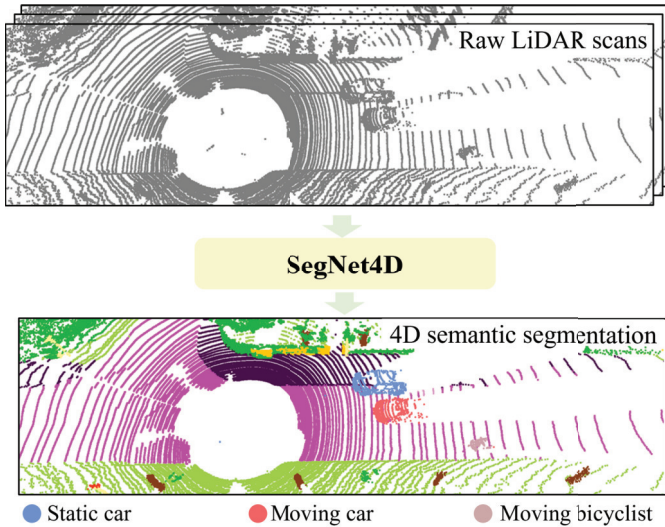


Fig. 1. Our SegNet4D performs real-time, online prediction of semantic categories while simultaneously identifying the motion state of each point. This capability greatly aids in downstream tasks including obstacle avoidance, path planning, and navigation, thereby enhancing the autonomous capabilities of robots.

segmentation (MOS). We argue that semantic classes and motion characteristics represent distinct aspects of objects. By addressing them separately and then fusing their outputs, our method achieves a more comprehensive understanding and superior performance compared to end-to-end approaches. To enhance efficiency, our framework converts the sequential LiDAR scans into Bird's Eye View (BEV) images and extracts motion features by calculating BEV residuals, significantly reducing computational cost compared to existing fashion using 4D convolutions or recursive networks. Furthermore, we find instance information [17], [18] is crucial for the LiDAR segmentation tasks to prevent over-segmentation. However, existing methods struggle to integrate this information effectively. To address this, we incorporate instance consistency from the current scan into the prediction pipeline at both feature and point levels, enabling instance-aware segmentation. Finally, we design a novel module to fuse point-wise semantic predictions and motion states to achieve accurate 4D semantic segmentation online. We extensively evaluate SegNet4D on mainstream datasets, comparing its performance with state-of-the-art (SOTA) methods. The results confirm that our approach surpasses SOTA in both 4D semantic segmentation and MOS. Through detailed runtime analysis, our method shows greater efficiency, enabling real-time operation. Additionally, we validate its practical utility on a real robotic platform, demonstrating both its effectiveness and efficiency for autonomous applications.

In summary, the contributions of this work are threefold:

- We propose a novel 4D semantic segmentation framework that decomposes the task into two subtasks: Single-Scan Semantic Segmentation (SSS) and Moving Object Segmentation (MOS). The results are then combined using a motion-semantic fusion module, achieving SOTA performance in both 4D semantic segmentation and MOS.
- We propose a novel instance-aware backbone that leverages instance information at both the feature and point

levels, enhancing performance in both MOS and semantic segmentation tasks. Additionally, we introduce an automated method to generate instance bounding box labels directly from semantic annotations, eliminating the need for additional manual labeling.

- We propose an efficient 4D semantic segmentation network that is significantly faster than existing methods and supports real-time operation. Furthermore, we successfully integrate the network into a real platform, demonstrating its practical utility for autonomous applications.

This article is an extension of our previous conference paper, i.e., InsMOS [17], which proposed to add instance information for MOS. Compared to [17], this article extends it in four critical aspects:

- a fully 4D semantic segmentation method that predicts both the motion states and semantic categories.
- a novel module for integrating point-wise motion states and semantic category predictions.
- a more efficient mechanism for encoding motion features, enabling this method to operate in real-time.
- a more extensive experimental evaluation on multiple datasets and a real-world unmanned ground platform, demonstrating this method's superior performance and practical utility.

II. RELATED WORK

A. Moving Object Segmentation

MOS refers to accurately identifying dynamic regions or objects from input data. The task has attracted extensive research in the field of computer vision [19], [20]. In recent years, some research has also been paid attention to LiDAR-based MOS. We categorize these methods into projection-based and non-projection-based.

The projection-based method typically converts sequential point clouds into image representations, such as range images [2], [21], [22], [23] or BEV images [24], [25], and then compute the residuals of sequential images to extract motion information in the scene. Chen et al. [21] first released the MOS datasets and benchmark based on the SemanticKITTI dataset [14], and proposed a deep neural network to learn motion cues from sequential residual range images for online MOS. To extract motion features more comprehensively, Sun et al. [22] proposed a dual-branch network architecture that separately encodes the appearance features and temporal motion features, and then fuses them with a multi-scale motion-guided attention module. Subsequently, Cheng et al. [23] utilized a similar dual-branch architecture and proposed a distribution-based data augmentation method to improve the network robustness. Kim et al. [2] also designed a similar multi-branch network to improve MOS accuracy, where one branch performs semantic segmentation to obtain movable objects, and another branch network extracts motion features. Finally, a fusion branch network is utilized to merge these features to achieve better MOS. Unlike range images, BEV images offer a top-down perspective, allowing a more intuitive observation of object movement. Based on this,

Mohapatra et al. [24] designed a lightweight network architecture for extracting motion information from BEV residual images and successfully achieved real-time operation in an embedded platform. However, this method can only perform pixel-wise segmentation in the BEV space and has relatively low accuracy. Recently, Zhou et al. [25] projected 3D point clouds into the polar coordinate BEV space and designed a dual-branch network structure similar to [22] to enhance the MOS performance.

Different from the projection-based methods, the non-projection methods directly extract motion information from sequential 3D point clouds. Mersch et al. [26] utilized the Minkowski engine [15] to construct a sparse 4D convolutional network. This network is designed to extract spatio-temporal features from the input 4D point clouds and predict point-wise moving labels. They also proposed a receding horizon strategy that integrates multiple observations to refine the network's predictions. Kreutz et al. [27] proposed an unsupervised MOS method based on [26], which learns spatio-temporal occupancy changes in the local neighborhood of point cloud videos. However, it is only applicable to stationary LiDAR. Due to the lack of instance-level perception, existing methods often partially segment moving objects. Therefore, we attempt to introduce instance information into the network, enabling it to achieve complete segmentation of moving objects.

B. 4D LiDAR Semantic Segmentation

Unlike the MOS task, 4D LiDAR semantic segmentation not only needs to capture temporal information to predict point-wise motion state but also assign a semantic category label to each point, including moving and static points. This substantially raises the challenge of the task by evolving a binary classification issue into a more intricate multi-class classification scenario, necessitating the network to acquire sophisticated appearance features and associate them with motion characteristics.

To address this task, some methods [1], [9], [14] try to fuse sequential LiDAR scans into a single point cloud and then utilize a 3D semantic segmentation network to predict 4D semantic labels directly. However, due to the lack of temporal perception capability in 3D networks, these methods often struggle with point-wise motion state recognition. Moreover, the fused single point cloud contains a substantial amount of data, greatly increasing the network's computation time and making real-time operation challenging. Consequently, such methods are typically not the preferred option in real-world scenarios.

Another category of method builds temporal-aware networks to tackle 4D semantic segmentation tasks. SpSequenceNet [11] employs two modules based on 3D convolutions, a cross-frame global attention module, and a cross-frame local interpolation module, to extract spatio-temporal information from input sequential LiDAR scans. TemporalLidarSeg [12] recursively extracts features from the sequential range image and utilizes a temporal memory alignment strategy to align the features of adjacent frames. Similarly, MemorySeg [28] also employs a features alignment strategy and operates in 3D space, which enhances point-level and voxel-level temporal

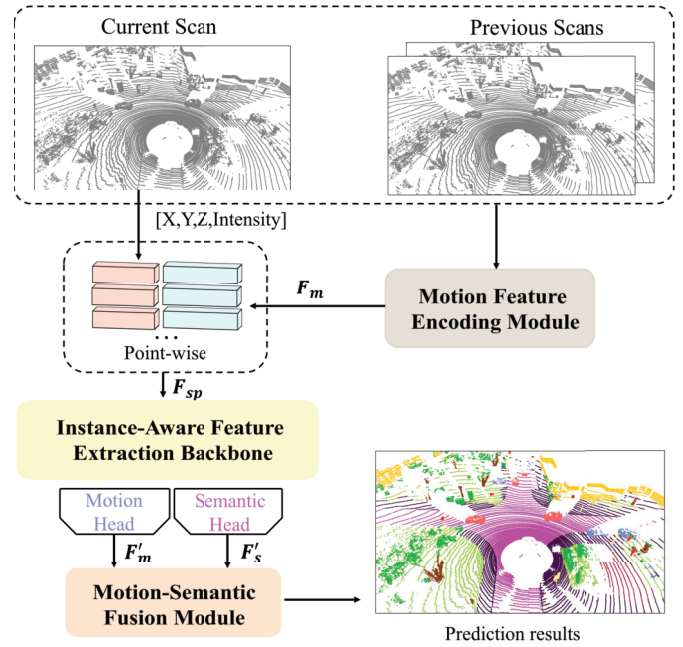


Fig. 2. The proposed framework of SegNet4D. We first utilize the Motion Features Encoding Module to extract motion features from the sequential LiDAR scans. Following this, the motion features are concatenated with the spatial features of the current scan and fed into the Instance-Aware Feature Extraction Backbone. Then, two separate heads are applied: a motion head for predicting moving states, and a semantic head for predicting semantic category. Finally, the Motion-Semantic Fusion Module integrates the motion and semantic features to achieve 4D semantic segmentation.

association. Schutt et al. [16] proposed a recurrent architecture for fusing the features of continuous point clouds and designed a module for gathering temporal information. However, the recursive extraction of features from point cloud sequences substantially amplifies the computational load on network processing. Recently, Liu et al. [13] proposed a module, MarS3D, which converts the sequential point clouds into BEV images and then utilizes an image network to extract motion features. This module can provide temporal awareness for existing 3D semantic segmentation networks. Shi et al. [29] propose a temporal variation-aware interpolation module and a temporal voxel-point refinement module to effectively model temporal dynamics within 4D point clouds. Besides, Chen et al. [30] proposed a sparse voxel-adjacent query network for processing temporal information, improving 4D semantic segmentation performance significantly.

For spatio-temporal feature extraction, some 4D panoptic segmentation methods [31], [32] also provide valuable references. Yilmaz et al. [31] propose spatio-temporal instance queries that encode the semantic and geometric properties of each semantic tracklet in the sequence, and they subsequently design an interactive 4D semantic segmentation network [33] based on this approach for dataset annotation.

The above-mentioned approaches require extracting temporally associated features from the sequential point clouds, resulting in time-consuming network operation. Although TemporalLidarSeg [12] and MarS3D [13] also utilize a projection-based fashion, they still rely on complex image processing networks to capture temporal features. In contrast, our method streamlines the process by directly incorporating

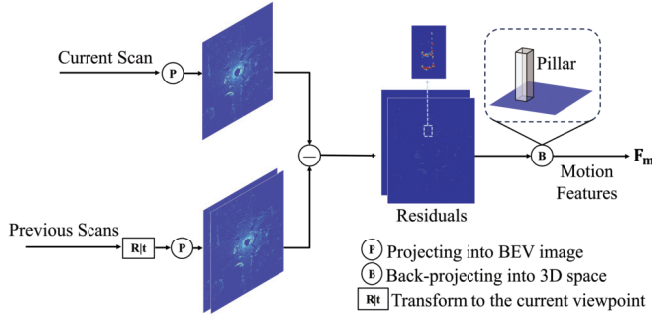


Fig. 3. The process for motion features encoding. We mainly calculate the residuals for the sequential BEV images and back-project them into the 3D space as the motion features.

pre-extracted motion residuals into the network as additional feature channels for network learning, which avoids intentionally extracting temporally associated features, thereby significantly boosting efficiency. Finally, by leveraging two separate output heads to supervise MOS and SSS explicitly and fusing them for 4D semantic segmentation, our method maintains superior segmentation accuracy.

III. METHODOLOGY

The framework of our proposed method is depicted in Fig. 2. SegNet4D consists of four main components: Motion Feature Encoding Module (Sec. III-A), Instance-Aware Feature Extraction Backbone (Sec. III-B), two separate output heads (Sec. III-C), and Motion-Semantic Fusion Module (MSFM, Sec. III-D). We specifically introduce each module in the following section.

A. Motion Feature Encoding Module

4D semantic segmentation not only predicts the semantic category for each point measured by LiDAR but also identifies its motion state. Therefore, it is necessary to extract motion features from the sequential point clouds. Existing methods mainly utilize 4D convolution [26], [27] to obtain motion cues, which poses a significant computational workload. To improve the real-time performance, we adopt the BEV image representation for fast motion feature encoding, as shown in Fig. 3, which avoids computational expense processing on extensive unstructured point cloud data. The encoding process is divided into three steps as follows.

1) *Point Cloud Alignment*: To compensate for the ego-motion of LiDAR, spatial alignment is conducted initially on the input sequential point clouds. Specifically, given the current point cloud $S_0 = \{p_i \in \mathbb{R}^4\}_{i=1}^M$ consisting of M points represented in homogeneous coordinates as $p_i = [x_i, y_i, z_i, 1]^T$, along with the past $N-1$ consecutive point clouds $S_1, S_2, \dots, S_{N-1}$ with their relative transformations $T_1^0, T_2^0, \dots, T_{N-1}^0$, we transform the past $N-1$ consecutive point clouds into the current viewpoint by

$$S_{j \rightarrow 0} = \{p'_i = T_j^0 p_i | p_i \in S_j\}, T_j^0 = \prod_{k=0}^{j-1} T_{j-k}^{j-k-1}. \quad (1)$$

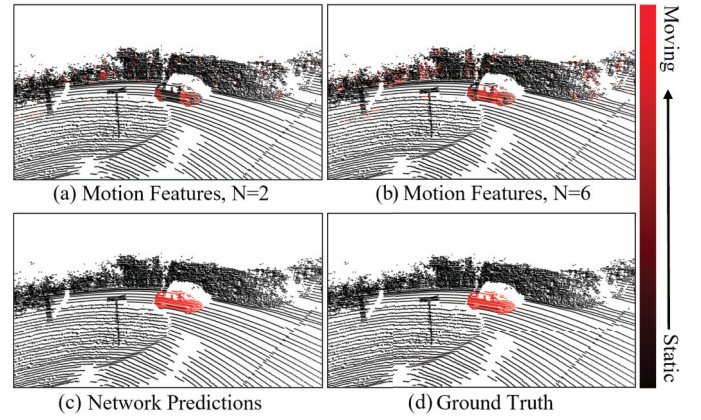


Fig. 4. Motion features visualization. (a) and (b) represent the motion features obtained from the current and past N -th scan. We compare the features with the network's predictions as well as ground truth.

In practical applications, the relative transformations can be easily obtained through the existing LiDAR odometry approach [4], [7]. We use the poses estimated by SuMa [7].

2) *BEV Projection*: After alignment, we project the aligned point clouds into single-channel BEV images. For each point $p'_j = (x'_j, y'_j, z'_j) \in S_{j \rightarrow 0}$, we first restrict it within $x'_j \in [X_{min}, X_{max}]$, $y'_j \in [Y_{min}, Y_{max}]$, $z'_j \in [Z_{min}, Z_{max}]$ and then convert it into the pillar space, given by

$$\begin{cases} l_{(u,v),j} = \{z'_j | z'_j \in p'_j\}, \\ u = \lfloor \frac{x'_j - X_{min}}{g} \rfloor, \\ v = \lfloor \frac{y'_j - Y_{min}}{g} \rfloor, \end{cases} \quad (2)$$

where $l_{(u,v),j}$ stores a set of point's height in the pillar located at (u, v) , and g denotes grid resolution.

Following [25], we project the l_j into a single-channel BEV image B_j of size $H \times W$. For each pixel value $B_{(u,v),j}$, we use the difference between the maximum and minimum height within the same pillar, calculating as:

$$B_{(u,v),j} = \text{Max}\{l_{(u,v),j}\} - \text{Min}\{l_{(u,v),j}\} \quad (3)$$

3) *Motion Features Encoding*: We take the BEV residuals $R \in \mathbb{R}^{H \times W \times (N-1)}$ between B_0 and $B_1, \dots, B_{N-1}$ as the motion features in the BEV space, calculated by

$$R_{(u,v),j \rightarrow 0} = B_{(u,v),0} - B_{(u,v),j}, j = 1, \dots, N-1, \quad (4)$$

where $R_{(u,v)}$ represents the residual value for pixel at (u, v) . To obtain motion features for each point in the 3D space, we perform back-projection by assigning the BEV residual value to all points projected to the BEV pixel. Points within the same pillar will share the same residual value for each residual image. Finally, we obtain point-wise motion features $F_m \in \mathbb{R}^{M \times (N-1)}$. We concatenate the F_m and current scan's spatial features, i.e., $[x, y, z, \text{intensity}]$, generating a new features $F_{sp} \in \mathbb{R}^{M \times (N+3)}$ as the input for subsequent backbone.

The extracted motion features visualization is presented in Fig. 4. Our approach can obtain initial motion cues for

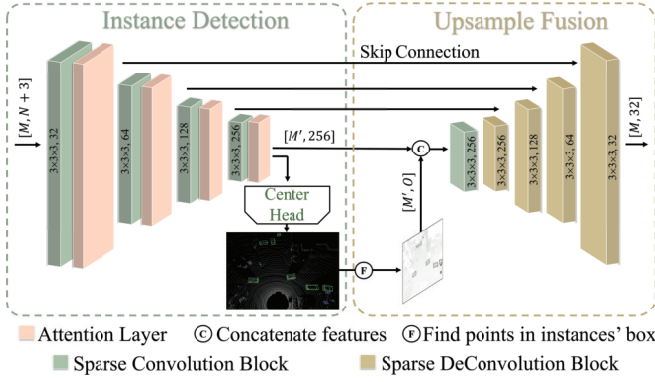


Fig. 5. The architecture of the Instance-Aware Feature Extraction Backbone. $3 \times 3 \times 3$ indicates a kernel size for 3D sparse convolution. $[32, 64, 128, 256]$ represents the dimensions of the output feature channels. M and N are defined in the motion feature encoding process. M' denotes the number of non-empty voxels in the last layer. O is the number of object detection categories.

moving objects in the scene. As the time interval for residual computation extends, the motion features become increasingly prominent. Inevitably, some points might be erroneously assigned motion characteristics because of changes in LiDAR viewpoint or inaccuracies in odometry estimation. However, through our network's subsequent learning process, which utilizes the scene's spatial features to distinguish movable from immovable objects and learning multi-channel motion features to identify true moving objects, these points still are correctly categorized, as shown in the Fig. 4 (c). Our network is capable of learning spatial features for semantic classification from the current LiDAR scan, as well as motion cues for motion segmentation. This is also a significant difference from purely projection-based methods, as we still leverage valuable spatial features inherent in the original point cloud.

B. Instance-Aware Feature Extraction Backbone

In semantic segmentation, over-segmentation is a common issue, where a single instance is incorrectly divided into multiple segments. Especially in the MOS task, existing methods lack instance awareness, often erroneously dividing an instance into two different motion states. To tackle such issues, we design the Instance-Aware Feature Extraction Backbone to extract instances information and incorporate them into our prediction pipeline. As shown in Fig. 5. It consists of the Instance Detection Module and the Upsample Fusion Module.

1) *Instance Detection Module*: In this module, We use 3D sparse convolution blocks [34] to voxelize the current point cloud and then perform convolution only on non-empty voxels, greatly improving the processing speed. Moreover, to capture more crucial features, we add a window self-attention [35] layer after each sparse convolution block to improve the receptive field. Our key finding is that instance information is crucial for LiDAR segmentation, as also shown in panoptic segmentation methods [36], [37]. However, these methods typically require additional point-wise instance labels, which may not always be accessible. To address this, we propose a method to automatically generate instance bounding boxes using existing semantic annotations. This process begins by

generating each instance cluster from the semantic point cloud using the Euclidean clustering algorithm in PCL library [38]. Subsequently, these clusters are projected into the BEV space, and their orientation is determined using principal component analysis (PCA) [39]. We then combine the height of the cluster to obtain the minimum bounding box and further refine it by L-Shape method [40]. Finally, we utilize CenterHead [41] trained with automatically generated instance labels, to predict instance bounding boxes of O categories online, such as cars, pedestrians, and cyclists. These predicted boxes are then used to enhance feature extraction within our backbone, as described in the upsample fusion module.

2) *Upsample Fusion Module*: To introduce instance consistency into feature-level fusion, we find and extract points within the instance bounding box as instance feature masks. Subsequently, we concatenate it with the point embeddings from the Instance Detection Module as input for the Upsample Fusion Module. Unlike InsMOS [17], we integrate the instance features only once to enhance efficiency. The Upsample Fusion Module serves the main purpose of integrating instance features into the prediction pipeline, as well as recovering the scale of the features hierarchically through 3D deconvolution blocks. It is formed by a fully U-Net network structure together with the Instance Detection Module and maintains more details from different scale features through skip connection.

C. Motion Head and Semantic Head

Existing 4D semantic segmentation methods usually predict all semantic class labels in an end-to-end manner, including those moving and static classes. However, since static points typically outnumber moving ones in existing datasets, these approaches often result in suboptimal network performance in identifying moving classes. So we employ two distinct heads for predicting moving labels and single-scan semantic labels separately. By explicitly supervising MOS, our method can maintain superior performance for moving object recognition.

To maintain the network's lightweight characteristics, these heads consist solely of a convolutional layer, a normalization layer, an activation function layer, and a linear layer, which are employed to further classify features with different attributes. Finally, we can obtain point-wise motion features $F'_m \in \mathbb{R}^{M \times 16}$ and semantic features $F'_s \in \mathbb{R}^{M \times 32}$, which are utilized for the subsequent fusion to enable 4D semantic segmentation. Where 16 and 32 denotes the number of feature channel. Naturally, the motion head and semantic head also generate point-wise motion predictions $F''_m \in \mathbb{R}^{M \times 3}$ and semantic predictions $F''_s \in \mathbb{R}^{M \times C}$ by applying an additional linear layer and a softmax function. Note that 3 represents three different motion classes, namely unlabeled, static, and moving, while C denotes the number of static semantic categories. Each head is supervised with a specific loss function. Further details are provided in Sec. III-F.

D. Motion-Semantic Fusion Module

After obtaining motion labels and single-scan semantic labels, a straightforward approach to achieve 4D semantic

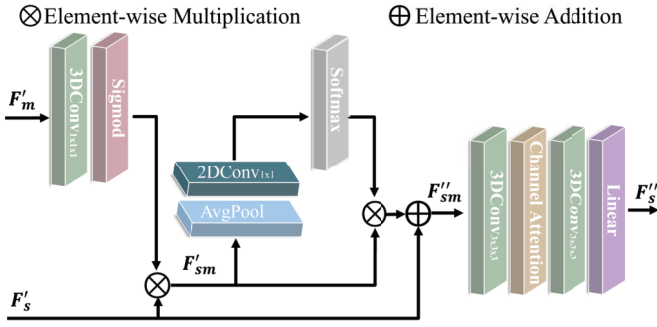


Fig. 6. The architecture of Motion-Semantic Fusion Module. We mainly perform spatial attention and channel attention to fuse the motion features and the static semantic features.

segmentation is to fuse them by checking the motion states on each semantic point. However, this fashion may yield non-smooth point segmentation. Moreover, incorrect motion predictions directly lead to false 4D semantic segmentation results. We hence propose a motion-semantic fusion module to further integrate motion features F'_m and static semantic features F'_s for achieving motion-guided 4D semantic segmentation. Details of this module are illustrated in Fig. 6.

We build our Motion-Semantic Fusion Module upon recently advanced motion-guided attention module [42] in the field of image processing. We extend it into 3D space and mainly utilize the 3D submanifold sparse convolution [34] as the backbone for performing spatial attention between the F'_s with F'_m as:

$$F'_{sm} = F'_s \otimes \text{Sigmoid}(3D\text{Conv}_{1 \times 1 \times 1}(F'_m)), \quad (5)$$

where $\otimes$ represent element-wise multiplication, $3D\text{Conv}_{1 \times 1 \times 1}(\cdot)$ represent a 3D submanifold sparse convolution with $1 \times 1 \times 1$ kernel size, and $F'_{sm} \in \mathbb{R}^{M \times D}$ are the fused motion-salient features, and D denotes the number of feature channels. We then perform channel attention for F'_{sm} to strengthen the responses of key attributes. Subsequently, it is element-wisely added by F'_s because the static semantic features are equally crucial for the 4D semantic segmentation. The final F''_{sm} is calculated as:

$$F''_{sm} = F'_{sm} \otimes [\text{Softmax}(2D\text{Conv}_{1 \times 1}(\text{AvgPool}(F'_{sm}))) \cdot D] + F'_s, \quad (6)$$

where $2D\text{Conv}_{1 \times 1}(\cdot)$ and $\text{AvgPool}(\cdot)$ denote a 2D convolution with 1×1 kernel size and average pooling operation, respectively. 2D convolution is mainly used to quickly calculate channel attention weights, which is different from using 3D convolution for spatial attention as mentioned earlier.

Finally, we further refine F''_{sm} using 3D sparse convolutions and channel attention blocks [43] to generate the final 4D semantic segmentation predictions $F''_s \in \mathbb{R}^{M \times C'}$, where C' is the number of categories for 4D semantic segmentation.

E. Moving Instance Refinement

To further enhance the accuracy of MOS, we check again the point-wise predictions within the instance bounding box and propose an instance-aware post-processing algorithm for

point-level refinement, combining in a bottom-up and top-down fashion.

This bottom-up aims to refine the results of the following second cases. Firstly, if many points within an instance are moving, then the instance is considered to be moving. Secondly, if the scene contains many moving vehicles, it is considered as a highly dynamic scene, such as the highway. Vehicles in this scene will be more easily classified as the moving class with a lower confidence threshold. During the top-down step, when an instance is identified as moving, all points within the instance are determined as moving, which is natural when considering each instance as a rigid body. Besides, because motion is a continuous process, we consider an instance to be moving only when classified as a moving class in multiple observations. The details for the algorithmic process please refer to our previous conference paper [17].

F. Loss Function

Due to multi-task setup, our loss function includes instance detection loss L_{det} , MOS loss L_{mos} , SSS loss $L_{\text{s_sem}}$ and 4D semantic segmentation loss L_{4D_sem} . In order to achieve a balance between the magnitudes of different loss functions and to expedite the convergence speed, we apply a weighted multi-task loss [44] L_{total} to supervise the training, defined as:

$$L_{\text{total}} = \sum_{i \in \{\text{det}, \text{mos}, \text{s_sem}, 4D_sem\}} \frac{1}{2\sigma_i^2} L_i + \ln(1 + \sigma_i^2), \quad (7)$$

where σ_i is a learnable parameter used to represent the uncertainty of L_i . L_{det} is composed of instance classification loss and bounding box regression loss. More details can be found in the CenterHead [41]. In addition, we employ the widely used weighed Cross-Entropy Loss function [22] for L_{mos} , $L_{\text{s_sem}}$ and L_{4D_sem} :

$$L_{\{\text{mos}, \text{s_sem}, \text{m_sem}\}}(y, \hat{y}) = - \sum \alpha_i p(y_i) \log(p(\hat{y}_i)), \alpha_i = 1 / \sqrt{f_i}, \quad (8)$$

where y_i and $\hat{y}_i$ denote the ground truth and the predicted labels, respectively. f_i is the frequency of the i -th class.

IV. EXPERIMENTAL EVALUATION

In this section, we conduct a series of experiments on SemanticKITTI [14] and nuScenes [45] datasets to demonstrate SegNet4D's capabilities on 4D semantic segmentation (Sec. IV-A) and MOS (Sec. IV-B) and compare the performance with SOTA. Subsequently, we integrate the method into a real robotic platform (Sec. IV-C), highlighting its practical utility. Besides, we perform ablation experiments to evaluate the effectiveness of the framework and the proposed modules (Sec. IV-D). Finally, we also conduct a detailed runtime analysis, demonstrating the high efficiency of our method (Sec. IV-E). These experimental results will substantiate our claims regarding the contributions.

1) *Datasets-SemanticKITTI* [14]: SemanticKITTI provides semantic labels for each individual LiDAR scan in the KITTI [46] odometry dataset, which comprises 22 sequences collected with a Velodyne HDL-64E LiDAR. Following previous

TABLE I
4D SEMANTIC SEGMENTATION PERFORMANCE EVALUATION ON THE SEMANTICKITTI BENCHMARK (MULTI-SCAN PHASE).
“MOV.” DENOTES MOVING. THE BEST RESULTS ARE IN BOLD

Methods	car	bicycle	motorcycle	truck	other-vehicle	person	bicyclist	motorcyclist	road	parking	sidewalk	other-ground	building	fence	vegetation	trunk	terrain	pole	traffic sign	mov. car	mov. truck	mov. other veh.	mov. person	mov. bicyclist	mov. motorcyc.	mIoU [%]
SpSequenceNet [11]	88.5	24.0	26.2	29.2	22.7	6.3	0.0	0.0	90.1	57.6	73.9	27.1	91.2	66.8	84.0	66.0	65.7	50.8	48.7	53.2	0.1	2.3	26.2	41.2	36.2	43.1
TemporalLidarSeg [12]	92.1	47.7	40.9	39.2	35.0	14.4	0.0	0.0	91.8	59.6	75.8	23.2	89.8	63.8	82.3	62.5	64.7	52.6	60.4	68.2	2.1	12.4	40.4	42.8	12.9	47.0
KPConv [9]	93.7	44.9	47.2	42.5	38.6	21.6	0.0	0.0	86.5	58.4	70.5	26.7	90.8	64.5	84.6	70.3	66.0	57.0	53.9	69.4	5.8	4.7	67.5	67.4	47.2	51.2
Cylinder3D [1]	93.8	67.6	63.3	41.2	37.6	12.9	0.1	0.1	90.4	66.3	74.9	32.1	92.4	65.8	85.4	72.8	68.1	62.6	61.3	68.1	0.0	0.1	63.1	60.0	0.4	51.5
MarS3D [13]	95.1	49.2	49.5	39.7	36.6	16.2	1.2	0.0	89.9	66.8	74.3	26.4	92.1	68.2	86.0	72.1	70.5	62.8	64.8	78.4	5.1	10.0	58.0	67.3	36.3	52.7
MemorySeg [28]	94.0	68.3	68.8	51.3	40.9	27.0	0.3	2.8	89.9	64.3	74.8	29.2	92.2	69.3	84.8	75.1	70.1	65.5	68.5	71.7	13.6	15.1	71.7	74.4	73.9	58.3
SVQNet [30]	96.1	64.4	60.3	40.4	60.9	27.4	0.0	0.0	93.2	71.6	80.5	37.0	93.7	72.6	87.3	76.7	72.3	68.4	71.0	80.5	3.9	7.5	84.7	72.4	91.0	60.5
Ours	95.1	60.7	59.1	54.5	55.3	17.8	19.3	31.0	91.6	64.9	76.4	35.2	93.2	71.6	85.7	72.5	70.1	62.7	69.5	84.8	11.2	30.5	60.6	72.5	76.3	60.9

work [1], [28], we use the standard data split where sequences 00 to 10 are used for training (with sequence 08 for validation), and sequences 11 to 21 for testing. The dataset contains multiple semantic category labels, and the semantic segmentation task is officially divided into two phases. One is the single-scan phase training 19 semantic classes without distinguishing point-wise dynamic attributes. The other is the multi-scan phase, which includes 25 semantic categories that distinguish between moving and static objects. To evaluate the 4D semantic segmentation performance of our SegNet4D, we test it in the multi-scan phase. In our model, the training is supervised with 26 semantic categories, including 6 moving classes, 19 static classes, and one *outlier* class. For the MOS task, All the semantic categories are reorganized into two classes: moving and static [21], [26]. Due to the imbalanced distribution of moving objects between the training and test set, an additional dataset, KITTI-Road, is introduced in [22] to mitigate the impact. We follow the experimental setups described in [22] to test the MOS performance.

2) *Datasets-nuScenes* [45]: It consists of 1000 driving scenes collected with a 32-beam LiDAR sensor. For point cloud segmentation tasks, it is mainly used for evaluating SSS with 16 static semantic classes. To achieve 4D segmentation evaluation, inspired by [13], we utilize the annotated bounding box motion attributes to generate 8 new moving categories, including *moving car*, *moving bus*, *moving truck*, *moving construction vehicle*, *moving trailer*, *moving motorcyclist*, *moving bicyclist*, and *moving person*. This is a new multi-scan phase similar to SemanticKITTI. We have released the multi-scan semantic segmentation dataset of nuScenes for the convenience of the community. For evaluating MOS, we divide all semantic categories into two classes (moving and static).

3) *Implementation Details*: We restrict the point cloud range in $[x : \pm 60m, y : \pm 50m, z : -4m \sim 2m]$ for SemanticKITTI, $[x : \pm 50m, y : \pm 50m, z : -4m \sim 2m]$ for nuScenes, and set $g = 0.1m$ for encoding motion features. For 4D semantic segmentation, we set $C = 20, C' = 26, O = 3$ for SemanticKITTI and $C = 17, C' = 25, O = 10$ for nuScenes. The proposed model is built on the PyTorch [47] library and

trained with 4 NVIDIA RTX 3090 GPUs. We set the batch size to 8 on a single GPU and train the network for a total of 80 epochs. The learning rate is initialized as 10^{-4} in the Adam optimizer [48] and a decay factor of 0.01 for each epoch. During the training process, we employ widely-used data augmentation techniques such as random flipping, scaling, and rotation to improve model performance.

4) *Evaluation Metrics*: For MOS performance evaluation, we use the Intersection-over-Union (IoU) [49] of moving objects as the metric:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}, \quad (9)$$

where TP, FP, and FN represent the predictions of the moving class that are classified as true positive, false positive, and false negative, respectively. For 4D semantic segmentation, we use the mean Intersection-over-Union (mIoU) across all categories as the evaluation metric.

A. Evaluation for 4D Semantic Segmentation

We evaluate the 4D semantic segmentation performance of our approach on the SemanticKITTI multi-scan semantic segmentation benchmark and nuScenes dataset, and compare the results with LiDAR-only SOTA baselines, including (a) single-scan-based methods (stack historical LiDAR scans into a single point cloud as input for multi-scans semantic segmentation): KPConv [9], Cylinder3D [1]; and specially designed multi-scans semantic segmentation methods: SpSequenceNet [11], TemporalLidarSeg [12], MarS3D [13], SVQNet [30] and MemorySeg [28]. Like all methods, we only utilize the past two LiDAR scans to predict semantic labels for a fair comparison, i.e., $N = 3$.

The quantitative results are presented in Tab. I. Our approach achieves a mIoU of 60.9% on the SemanticKITTI dataset and outperforms all methods, demonstrating its effectiveness. For semantic class *bicyclist* and *motorcyclist*, our SegNet4D obtains a significant improvement, indicating that the incorporation of instance information enables our model to identify these foreground points more effectively. Besides, our method boasts the fastest execution speed and is the only approach capable of real-time operation on the SemanticKITTI

TABLE II
4D SEMANTIC SEGMENTATION PERFORMANCE EVALUATION ON THE nuSCENES DATASET. “MOV.” DENOTES MOVING

Methods	barrier	bicycle	bus	car	construction	motorcycle	pedestrian	traffic cone	trailer	truck	driveable	other flat	sidewalk	terrain	manmade	vegetation	mov. car	mov. bus	mov. truck	mov. const.	mov. trailer	mov. motor.	mov. bicyc.	mov. person	mIoU [%]
SpSequenceNet [11]	65.3	0.0	0.0	0.0	23.2	0.0	52.2	39.2	26.3	52.6	94.5	67.5	67.6	70.5	82.1	81.3	46.0	33.4	26.0	0.0	18.9	38.9	0.0	52.5	39.1
KPConv [9]	65.0	32.0	58.3	59.3	39.4	37.4	27.6	49.3	29.9	61.8	79.3	55.7	60.0	55.5	44.6	41.8	54.3	49.5	47.9	0.0	42.3	68.9	39.4	57.6	48.2
Cylinder3D [1]	70.0	22.3	41.6	74.0	40.8	22.3	43.3	59.3	46.1	62.3	95.2	64.8	70.2	71.3	87.7	86.0	56.6	43.5	40.0	0.0	30.3	65.3	22.3	63.9	53.3
MarS3D [13]	70.5	24.7	60.0	79.9	32.0	34.9	51.3	53.0	10.4	66.0	95.4	59.9	72.7	75.8	87.2	86.1	66.5	48.0	52.4	0.0	23.1	69.0	9.7	72.7	54.3
Ours	77.4	32.6	63.8	73.8	41.1	44.0	51.2	63.2	42.3	74.2	96.2	69.3	74.2	73.5	64.6	55.9	68.6	51.3	59.4	0.0	27.2	74.3	40.8	72.4	57.9

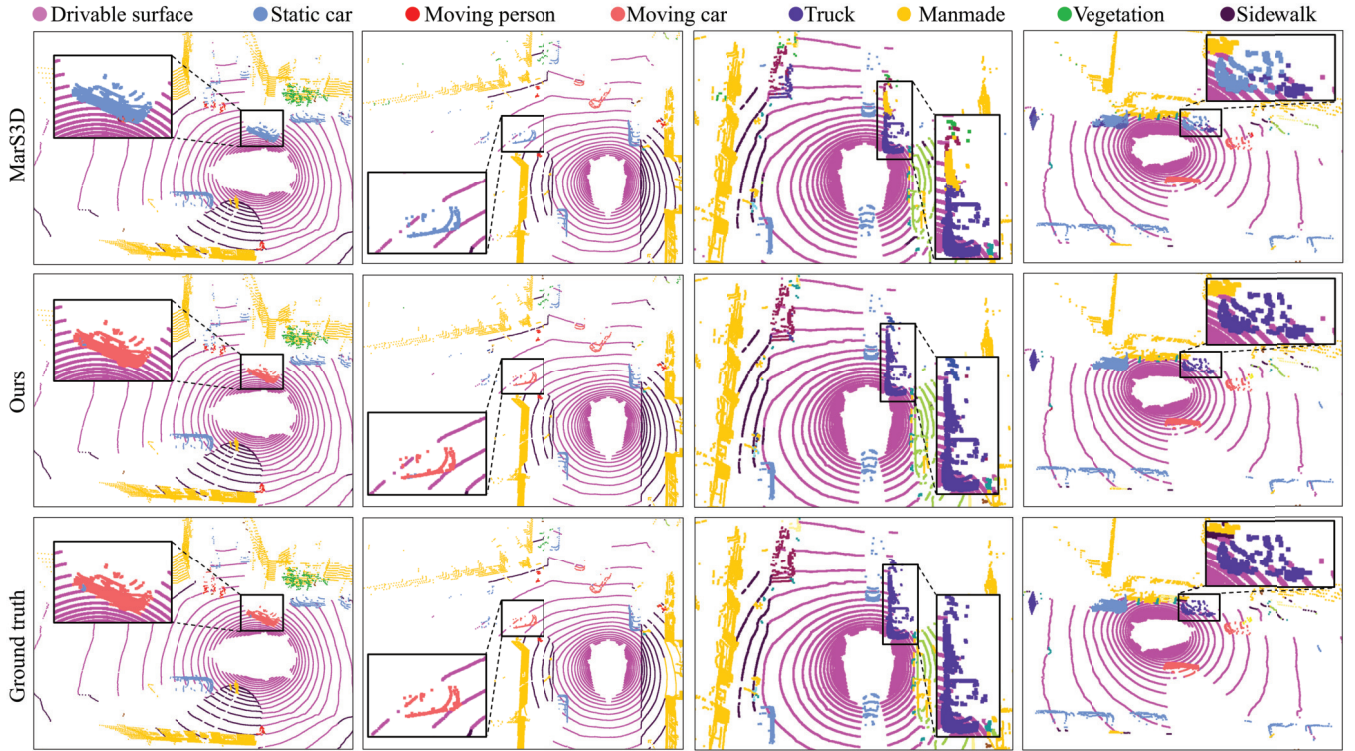


Fig. 7. The qualitative results comparison between our approach and MarS3D on the nuScenes dataset.

dataset, showing its highly efficient for 4D LiDAR semantic segmentation. For more runtime analysis, please refer to Sec. IV-E.

Additionally, we further evaluate our approach on the nuScenes datasets and compare the performance with other baselines. Note that some methods in the Tab. I is not open-source, preventing their evaluation in the nuScenes multi-scan phase. This also further highlights the contribution of our open-source release in supporting community research. As shown in Tab. II, our method consistently delivers the best performance, demonstrating its adaptability to different types of LiDAR and scene changes. The qualitative comparison with MarS3D, as illustrated in Fig. 7, reveals our approach’s superior ability for moving object recognition. This is also further corroborated by quantitative analyses presented in the Sec. IV-B. Additionally, our method exhibits complete segmentation for big instances, like trucks, contrasting with Mars3D’s incomplete segmentation results. This

contrast highlights the effectiveness of our instance-aware design.

B. Evaluation for Moving Object Segmentation

We evaluate the result on the SemanticKITTI-MOS benchmark and nuScenes dataset, and compare it with SOTA MOS methods, including (a) projection-based: LMNet [21], MotionSeg3D [22], RVMOS [2], MotionBEV [25] and MF-MOS [23]; (b) point-based: 4DMOS [26] and InsMOS [17]; as well as open-source 4D segmentation baselines: KPConv [9], SpSequenceNet [11], Cylinder3D [1] and MarS3D [13].

The quantitative comparison is presented in Tab. III. Our method achieves the best results on both the SemanticKITTI and nuScenes dataset, demonstrating its superior performance in motion segmentation. SegNet4D also shows improved performance compared to original InsMOS [17], indicating that semantic information is beneficial for the identification of moving objects, as it provides a vital cue for distinguishing

TABLE III
MOS PERFORMANCE EVALUATION (IoU [%])

Methods	Publication	Semantic KITTI	nuScenes
LMNet [21]	RAL 2021	62.5	49.9
4DMOS [26]	RAL 2022	65.2	67.8
MotionSeg3D [22]	IROS 2022	70.2	63.0
RVMOS [2]	RAL 2022	74.7	-
InsMOS [17]	IROS 2023	75.6	65.7
MotionBEV [25]	RAL 2023	75.8	64.7
MFMOS [23]	ICRA 2024	76.7	62.6
KPConv [9]	ICCV 2019	60.9	56.2
SpSequenceNet [11]	CVPR 2020	43.2	42.9
Cylinder3D [1]	TPAMI 2022	61.2	59.5
MarS3D [13]	CVPR 2023	66.2	64.3
Ours	-	78.5	69.0

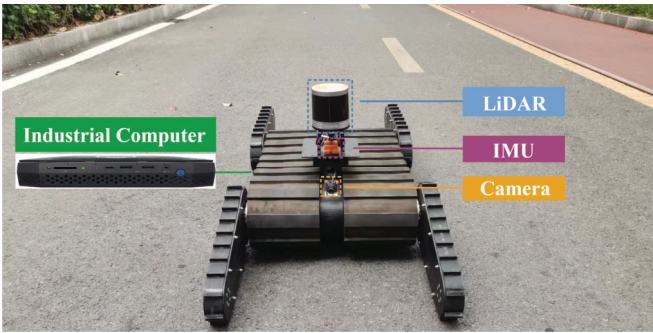


Fig. 8. Our self-developed tracked mobile platform for experiments. For perception tasks, such as segmentation or detection, there is no difference with respect to wheeled platforms.

between movable and immovable classes. Additionally, our method significantly outperforms other 4D semantic segmentation methods in terms of MOS performance, benefiting from our framework’s explicit MOS supervision during the training phase.

C. Operation on Real-World Platform

To substantiate the real-world utility of SegNet4D, we further simplify and accelerate the model using half-precision floating-point arithmetic and parallel motion feature encoding, subsequently integrating it into an operational robotic system. As shown in Fig. 8, our platform is equipped with a RoboSense Helios-32 LiDAR with 10 Hz sampling frequency, an Xsens MTI-300 IMU, a CMOS camera, and an industrial computer with Intel Core i7-1165G7, 16G RAM, and NVIDIA RTX 2060 GPU. In the operational phase, we utilize LiDAR and IMU data to calculate odometry poses using the FAST-LIO [50] algorithm.

Our method achieves real-time operation at 15.7Hz rate on the self-developed platform, faster than the typical rotation LiDAR sensor frame-rate of 10Hz. We also utilize this platform to collect a dataset from the campus scene and qualitatively compare the semantic segmentation results with Cylinder3D [1] and Mars3D [13]. All methods are only trained on the nuScenes dataset and directly applied for inference on our dataset. As illustrated in Fig. 9, we can see that our SegNet4D gets the best segmentation results, demonstrating its

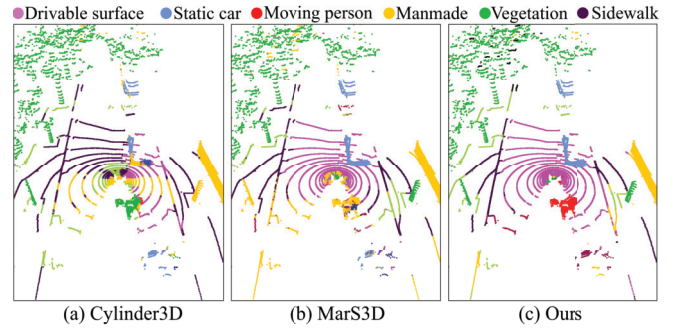


Fig. 9. The qualitative comparison on our campus scene. Our approach can more accurately identify the drivable surfaces, persons, and manmade buildings, whereas Cylinder3D and MarS3D show a decline in performance.

TABLE IV
MOS GENERALIZABILITY EVALUATION ON OUR CAMPUS DATASET

Methods	IoU[%]
Cylinder3D [1]	25.2
MarS3D [13]	40.4
Ours	49.8

TABLE V
ABLATION STUDIES ON THE nuSCENES VALIDATION SET. **MOS**: MOVING OBJECT SEGMENTATION. **SSS**: SINGLE-SCAN SEMANTIC SEGMENTATION. **MANUAL MS**: MANUALLY MERGING MOS AND SSS FOR MULTI-SCAN SEMANTIC SEGMENTATION. **NETWORK MS**: THE NETWORK’S PREDICTIONS FOR MULTI-SCAN SEMANTIC SEGMENTATION, I.E., THE OUTPUT OF MSFM FOR [A],[C],[D], THE OUTPUT BY ONE HEAD FOR [B]

	Inst- ance	One Head	Two Head	Refin- ement	MOS	SSS	Manual MS	Network MS
[A]			✓		67.1	67.3	55.5	56.3
[B]	✓	✓			-	-	-	56.0
[C]	✓		✓		68.5	68.6	57.4	57.9
[D]	✓		✓	✓	69.0	68.6	57.5	57.9

practical utility for the real platform. Additionally, we manually annotate a sequence (900 frames) with moving labels to quantitatively evaluate the MOS generalization performance. From the results shown in Tab. IV, our method still possesses the best generalization ability.

The above-mentioned experimental results show that SegNet4D can operate in real-time on the real unmanned ground platform while exhibiting superior performance in 4D semantic segmentation and MOS. This highlights its practical utility in enhancing robots’ environmental perception capabilities.

D. Ablation Study

In this section, we conduct a series of ablation experiments on our framework and MSFM module to test their effectiveness. The results are shown in Tab. V. Here, “Instance” indicates whether instance information is integrated into the Upsample Fusion Module to achieve instance-aware segmentation. “One Head” means directly predicting multi-scan semantic labels using only one head, indicating an end-to-end fashion. “Two Head” means utilizing two heads (the motion head and the semantic head, i.e., our framework) to predict moving objects and single-scan semantic labels, respectively,

TABLE VI
RUNTIME ANALYSIS. THE BEST RESULT ARE IN BOLD

Methods	Runtime [ms] (Sem.KITTI)	Runtime [ms] (nuScenes)	Params [$\times 10^6$]	FLOPs [$\times 10^9$]
SpSequenceNet [11]	497.3	475.6	6.8	104.3
KPConv [9]	139.6	95.6	18.4	99.8
Cylinder3D [1]	125.0	108.2	55.9	126.5
Mars3D [13]	180.5	129.4	21.9	175.6
Ours	67.1	53.6	35.7	71.1

and finally merging their results to achieve multi-scan semantic segmentation. “Refinement” refers to moving instance refinement mentioned in Sec. III-E, and it only focuses on refining the motion predictions.

By comparing [A] and [C], we can see that incorporating instance information can improve the accuracy of MOS and semantic segmentation, which equips the network with instance-aware segmentation capabilities, and the results substantiate our second claim. In the [B] and [C], predicting multi-scan semantic labels directly from a head is even inferior to the results obtained by manually integrating outputs from the motion head and the semantic head. This illustrates the superiority of our framework in comparison to directly predicting multi-scan semantic labels using an end-to-end manner, thereby supporting our first claims. The results of [C] and [D] indicate that the refinement is effective for improving the motion predictions, which further proves the utility of instance information. Besides, we manually merge the predictions of MOS and SSS, labeled as Manual MS in the Tab. V, and compare its results with the output of the proposed MSFM. We can see that the MSFM’s segmentation result always outperforms manual fusion, confirming its effectiveness in integrating motion and static semantic predictions.

E. Runtime Analysis

To thoroughly assess the computational efficiency of our proposed network, we conduct a comprehensive analysis of its inference time, parameter count, and floating-point operations (FLOPs) utilizing a single NVIDIA 3090 GPU, benchmarking against all open-source 4D semantic segmentation networks. As shown in Tab. VI, our method achieves the fastest runtime on both the SemanticKITTI and nuScenes datasets, while also exhibiting the lowest FLOPs, demonstrating its superior computational efficiency. Furthermore, our network comprises 35.7M parameters, indicating small memory consumption suitable for limited onboard resources.

V. CONCLUSION AND FUTURE WORK

In this paper, we present a novel 4D semantic segmentation method to predict both point-wise moving labels and semantic labels for LiDAR data, and operate in real-time. The framework decomposes the complex 4D semantic segmentation task into MOS and SSS tasks, finally merging their predictions to achieve more accurate 4D semantic segmentation. We adopt a projection-based approach to quickly obtain motion features, which significantly reduces the computational complexity

compared to 4D convolutions. To achieve instance-aware segmentation, we concatenate the motion features with the spatial features of the current scan, feeding them into the network for instance detection, and subsequently inject the instance features into the prediction pipeline. In addition, we design a motion-semantic fusion module to integrate the point-wise motion states and static semantic predictions, enabling motion-guided 4D semantic segmentation. Extensive experiments on multiple datasets and a real-world unmanned ground platform demonstrate the superiority of our method. The detailed runtime analysis further highlights its computational efficiency, confirming its real-time operation feasibility across diverse LiDAR configurations.

Despite its strengths, our system faces limitations, notably, BEV-based motion extraction fashion may overlook moving objects obscured in the Z-direction. Future studies may integrate the range images to enhance the robustness of motion feature extraction. Additionally, SegNet4D currently cannot operate on embedded GPU platforms due to limitations in 3D sparse convolution version. Subsequently, we may employ TensorRT to accelerate the model, further enhancing its efficiency and integrating it into low-cost robot platforms. Beyond the aforementioned, future studies may exploit the instance information we introduced to broaden the network’s capabilities, including panoptic segmentation [37] and 4D panoptic segmentation [36]. Furthermore, studies could leverage our predicted 4D semantic labels to boost robotic autonomy, particularly in applications such as semantic SLAM [6], semantic-based navigation [3] or planning [51].

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Neng Wang received the B.E. degree in automation from Southwest Petroleum University, China, in 2021. He is currently pursuing the Ph.D. degree with the National University of Defense Technology (NUDT). His research interests include point cloud segmentation and semantic-based simultaneous localization and mapping (SLAM).



Ruibin Guo received the B.S. degree in communication engineering, the M.S. degree in information and communication engineering, and the Ph.D. degree from the National University of Defense Technology (NUDT), in 2012, 2014, and 2019, respectively. He is currently a Lecturer with NUDT. His research interests include computer vision, simultaneous localization and mapping, and 3D reconstruction.



Chenghao Shi received the B.E. degree from Nanjing University of Aeronautics and Astronautics (NUAA) and the M.E. and Ph.D. degrees in robotics from the National University of Defense Technology (NUDT), China, in 2017, 2019, and 2024, respectively. He is currently a Lecturer with the NUDT, China. His research interests include localization and mobile robots.



Ziyue Wang received the Bachelor of Engineering degree from Nanjing University of Science and Technology (NJUST) in 2023. He is currently pursuing the Ph.D. degree with the National University of Defense Technology (NUDT). His research interests include uncertainty of mapping.



Hui Zhang received the B.E., M.E., and Ph.D. degrees from the National University of Defense Technology, China, in 1993, 1996, and 2000, respectively. He became a Professor with the National University of Defense Technology in 2011. His research interests include mobile robots, simultaneous localization and mapping, and intelligent control.



Huimin Lu (Member, IEEE) received the B.E. degree in automation and the M.E. and Ph.D. degrees in control science and engineering from the National University of Defense Technology (NUDT), Changsha, China, in 2003, 2005, and 2010, respectively. He joined the College of Intelligence Science and Technology, NUDT, in 2010, where he is currently a Professor. His research interests include mobile robotics, mainly robot vision, multi-robot coordination, and human-robot interaction.



Zhiqiang Zheng received the Ph.D. degree in aerospace engineering from the University of Liege, Liege, Belgium, in 1994. He is currently a Professor with the College of Intelligent Science and Technology, National University of Defense Technology, Changsha, China. His research interests include precision guidance and control and multirobot coordination control.



Xieyuanli Chen (Member, IEEE) received the bachelor's degree in electrical engineering and automation from Hunan University, China, in 2015, the master's degree in robotics from the National University of Defense Technology, China, in 2017, and the Ph.D. degree from the Photogrammetry and Robotics Laboratory, University of Bonn. He is currently an Associate Professor with the National University of Defense Technology, China. He also serves as an Associate Editor for IEEE ROBOTICS AND AUTOMATION LETTERS.